BREADBOARD

DIGITAL LOGIC SIMULATOR

# User Guide

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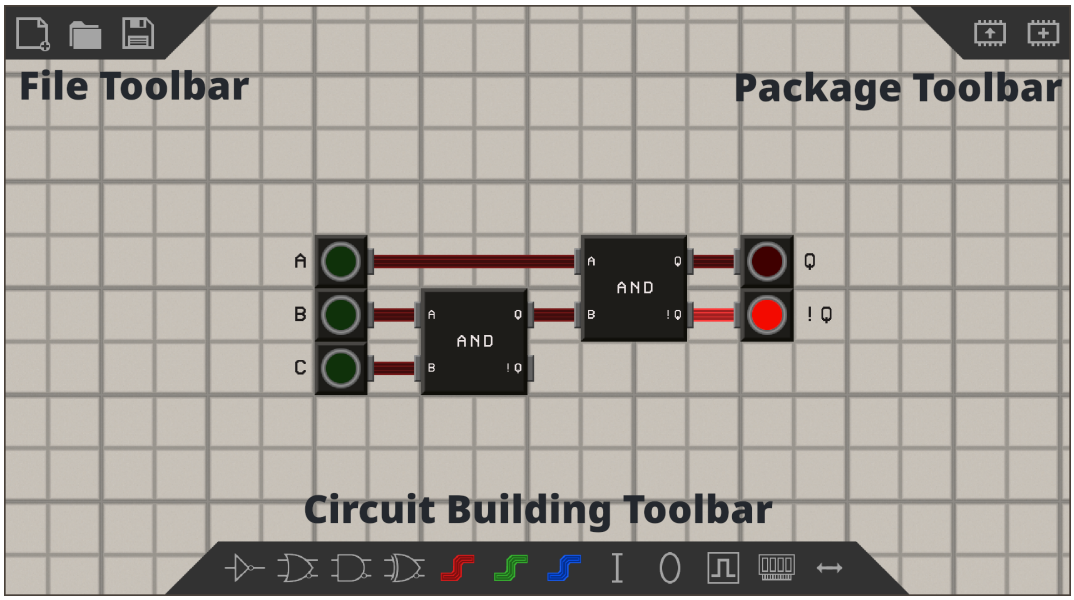
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# 1 Introduction

Breadboard is a digital circuit simulator, designed to be used as an educational tool, simple enough to be easy for a newcomer to digital logic to learn, but complete in its capacity to simulate large and complex circuits. It provides you with a number of useful features to make the creation of circuits simple and intuitive, such as building larger circuits from smaller circuit packages of your own design.




## 1.1 Circuit Board

The large tiled space displayed in the main portion of the screen is the circuit board, this is where you will place circuit packages and draw wires to build your circuits.

## 1.2 Toolbar

The toolbar, displayed at the bottom of the screen, provides you with a selection of tools for building circuits. (See section 3 for detailed information on what these buttons do.)

## 1.3 File Options

At the top-left of the screen is the file options toolbar, this gives you a few options for saving and loading your projects:

**New File:**

Opens a new, empty project, replacing the current one. You will be prompted to save your currect project before a new one is created.

**Open File:**

Opens a window with a scrollable menu from which you can select a project file. You can then use the buttons provided to either load or delete the selected file.

**Save File:**

Opens a text dialog which allows you to name and save your file to be loaded at a later time, or used as a custom circuit package.

## 1.4 Package Options

At the top-right of the screen is the package options toolbar, this gives you the options of saving and loading your own custom circuit packages. (See section 4 for more information.)

**Open Package:**

Opens a small window with a scrollable menu from which you can select a project file. Clicking this button again will close the window. Selecting a project file will create a new circuit package with the project file's contained circuit inside of it, the package pins will be based on the input and output nodes contained in the file. The pins will have the same name as its corresponding input/output node.

**Save Package:**

Opens a text dialog which allows you to name and save your package to be loaded at a later time. The circuitry saved in this file is determined by the selection box. (See section 2.4 for information on the selection box.)

# 2 Controls

## 2.1 Movement

You can move around the circuit board using the arrow keys or the W, A, S and D keys. Holding the SHIFT key will double the speed at which the camera moves. Using the scroll-wheel on your mouse, you can zoom in and out of the board. Movement speed is proportional to your zoom level, so zooming out will allow for faster navigation of the circuit board.

## 2.2 Wire Placement

Selecting the red, green or blue wire buttons on the toolbar will activate wire mode. Clicking and/or holding the left mouse button will allow you to draw wires of the selected colour on the circuit board. Wires of different colours will not connect to each other, allowing you to layer your wires. Clicking and/or holding the right mouse button will allow you to delete wires of the chosen colour. Clicking on the selected wire color again will disable wire mode.

Wires are displayed by the order of the most recently selected colour. Selecting a wire tool will bring that wire to the top, displaying it on top of other wires of different colours.

## 2.3 Package Placement

Selecting a package from the toolbar, loading a package from the “Open Package” menu, or grabbing circuitry using the selection box, will bind that circuitry to your mouse. You can

then place this package/circuitry using the left mouse button at the location you wish to place it on the circuit board.

### 2.4 Making Selections

When you are not holding any circuitry with your mouse, and not currently drawing any wires, you can hold down the left mouse button to draw a box. A left click inside this box will grab the circuitry enclosed within the box, allowing you to then place it somewhere else on the board.

**NOTE:** Packages must be contained entirely within the border of the selection box to be counted as being selected.

### 2.5 Copy and Paste

Drawing a selection box over some circuits and pressing CTRL+C will copy the selected circuits. CTRL+V will paste what you have previously copied to your mouse, ready to be placed on the board with the left mouse button.

### 2.6 Deletion

Pressing the DEL or ESC buttons will delete any wires or packages currently held by the mouse.

## 3 Wires and Circuits

### 3.1 Wires

Wires are used to connect circuit packages together by their pins. Wires carry signals from one pin to another. By default, wires are in the “off” state. When a wire receives a signal from one of the package pins it is connected to, it will switch to the “on” state, highlighting it to visually communicate its current state. Wires will pass their current state

to any package's input pins that they are connected to. When a wire is no longer recieving an "on" signal from any pins, the wire will default to the "off" state.

## 3.2 Circuit Packages

Circuit packages are the building blocks of circuit logic, they are a representation of some operation, either a simple logic gate such as AND or OR, or as a compacted version of a more complex circuit, reduced down to only its inputs and outputs. These inputs and outputs appear on the package as pins on either side. Pins on the left side of a package are inputs, in that they receive signals from wires, and pins on the right side of a package are outputs, sending signals to connected wires.

The following sections will detail the functions of circuit packages provided by the toolbar, located at the bottom of the screen.

### 3.2.1 Logic Gates

These packages represent the elementary logic functions used in building circuits:

**NOT Gate:**

This gate has one input and one output. Its output is the opposite of whatever signal it is receiving at its input.

**OR Gate:**

This gate has two inputs (A and B) and two outputs (Q and !Q). Q will switch to the "on" state when any of its inputs are in the "on" state. !Q is the negation of Q.

**AND Gate:**

This gate has two inputs (A and B) and two outputs (Q and !Q). Q will only switch to the "on" state when both A and B are in the "on" state.

**XOR Gate:**

This gate has two inputs (A and B) and two outputs (Q and !Q). Q will only switch to the “on” state when its inputs are in different states from each other.

### 3.2.2 Input/Output Nodes

These packages are used to input and display signals to and from connected wires. Input nodes will toggle their state when clicked, outputting it’s current state via it’s only pin. Output nodes will light up when given an “on” signal by a wire. These nodes are used when determining the names of pins in custom circuit packages. Input and output nodes can be renamed by right-clicking on them and entering a new name into the text box and clicking the “confirm” button.

### 3.2.3 Clock

This package will output an alternating signal at a constant rate. This rate is currently static and cannot be modified.

### 3.2.4 Memory (8-bit)

This package provides a fast, 8-bit memory block. Here are details on its input and output pins:

**Data Pins (Dx):**

These eight pins decide what data should be written to a memory cell.

**Address Pins (Ax):**

These eight pins select which memory cell should be written to or read from.

**Output Pins (Ox)**

These eight pins output the data stored at the selected memory address when the load pin is activated.

**Load Pin (LD):**

Tells the memory package to pass the contents of the selected memory cell to the output pins.

**Store Pin (ST):**

Tells the memory package to store the data on the data pins at the selected address when the clock pin is activated. (This will not store the data if the clock pin is already active when the store pin is activated.)

**Clock Pin (CLK):**

Synchronises storage of data to selected memory cell. Should be connected to clock.

### 3.2.5 Connector

This package's output is always equal to its input. It is intended to allow you to combine multiple wires to a single output, switch wire colours for wire organisation, or bridge the same wire colour over each other to prevent them from connecting, all in a compact fashion.

## 3.3 Error Detection

Breadboard is capable of detecting when a circuit or sub-circuit is oscillating. This is where a part of a circuit "flickers" and is unable to determine whether it should be in the "on" or "off" state. This can easily be demonstrated by connecting a NOT gate's output to its input. When this is detected, the simulation will halt until either an edit is made to the circuit, or an input node is toggled. Circuits that may be the source of this problem will be highlighted in red.

# 4 Custom Circuit Packages

Custom circuit packages are abstractions of a circuit you have built and saved, the details of the circuit's operation are hidden, and reduces the circuit to just its input and output nodes. This allows you to keep your circuits cleaner and more understandable in larger projects.

## 4.1 Making Packages

Using the selection box, you can select an area of the board containing the circuitry you wish to convert into a circuit package. Make sure you have all the input and output nodes in place, as they will be used to determine the package's input and output pins. It is recommended that you name your input and output nodes to distinguish them from each other when the package is created.

Once you have made your selection, click the "Save Package" button in the top-right corner of the screen, and enter the name of the package when prompted. This will convert the selected circuitry into a circuit package and save it to a file. Saved circuits can be accessed via the "Open Package" menu, the button for which is located at the top-right corner of the screen.

## 4.2 Editing Packages

You can edit existing packages by opening the package file via the "Open File" button at the top-left of the screen. You can edit and save your package here. Editing and saving a custom circuit package will not automatically replace any existing instances of your package in other files, these must be replaced manually.

## 4.3 Deleting Packages

Package files can be deleted through the "delete" option in the "Open File" menu. Deleting a package does not invalidate any existing instances of the package in other circuits, so

feel free to delete the files of any one-time packages you do not wish to keep.

# 5 Troubleshooting

## 5.1 Windows: Crash on first startup

This is most likely a problem with the way Windows handles file permissions and should not persist past the first time the program is run.

## 5.2 Windows: Anti-virus Warning

Because this program has not been officially signed with a certificate by a trusted authority, several anti-virus programs will determine this program to be unsafe due to “low reputation”. Simply ignore this error, or whitelist the program to continue.

## 5.3 Program crashes on startup multiple times

This can be caused by missing files, running the program from inside a compressed archive file or, more commonly, running the program from anywhere other than the root directory of the project (where the README.md file is located).